COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 031 - VDA	**SECTION:**	CPE21S1
DATE PERFORMED:	09/12/26	**DATE SUBMITTED:**	09/12/26
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ACTIVITY NO. 7 Purpose Use of Color

1. PROCEDURE OUTPUT

What procedures did you perform?

TASK 1 : COLOR AND PERCEPTION

Code Snippet

```
import matplotlib.pyplot as plt
import numpy as np

# Sample data for categorical example
categories = ["A", "B", "C", "D", "E"]
values = [10, 30, 20, 40, 25]

# Grid for sequential and diverging examples
x = np.linspace(-3, 3, 100)
y = np.linspace(-3, 3, 100)
X, Y = np.meshgrid(x, y)
Z_seq = np.sqrt(X**2 + Y**2)     # Sequential data (distance from origin)
Z_div = X                        # Diverging data (negative to positive)

# Create figure
fig, axes = plt.subplots(1, 3, figsize=(15, 5))

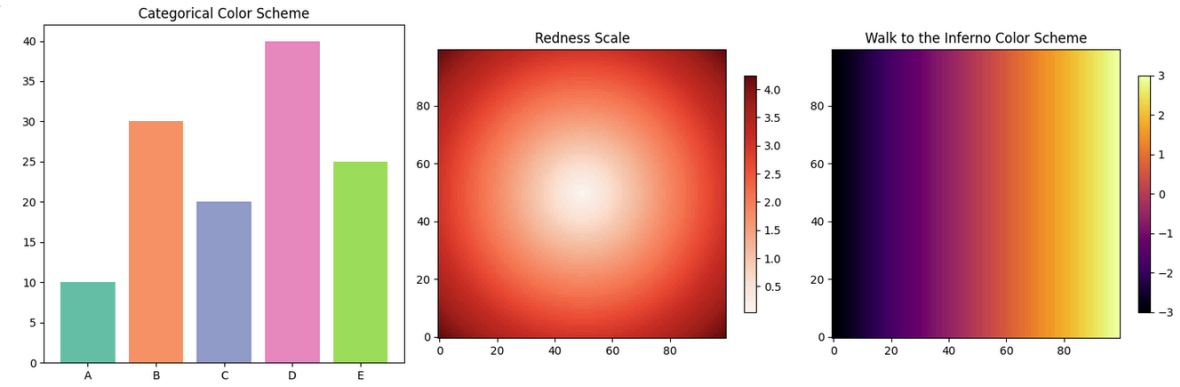
# ------------------------------
# 1. Categorical Color Scheme
# ------------------------------
axes[0].bar(categories, values, color=plt.cm.Set2.colors[:len(categories)])
axes[0].set_title("Categorical Color Scheme")

# ------------------------------
# 2. Sequential Color Scheme
# ------------------------------
seq = axes[1].imshow(Z_seq, cmap="Reds", origin="lower")
axes[1].set_title("Redness Scale")
plt.colorbar(seq, ax=axes[1], shrink=0.7)

# ------------------------------
# 3. Diverging Color Scheme
# ------------------------------
div = axes[2].imshow(Z_div, cmap="inferno", origin="lower")
axes[2].set_title("Walk to the Inferno Color Scheme")
plt.colorbar(div, ax=axes[2], shrink=0.7)

# Show results
plt.tight_layout()
plt.show()
```

Output Snippet



TASK 2 : COLOR SCHEMES AND TYPES OF DATA

Code Snippet

```
import matplotlib.pyplot as plt
import numpy as np

categories = ["Drinks", "Fruits", "Accessories", "Toiletries", "Vegetables"]
values = [39, 25, 12, 26, 19]

x = np.linspace(-3, 3, 100)
y = np.linspace(-3, 3, 100)
X, Y = np.meshgrid(x, y)
Z_seq = np.sqrt(X**2 + Y**2)
Z_div = X

fig, axes = plt.subplots(1, 3, figsize=(15, 5))

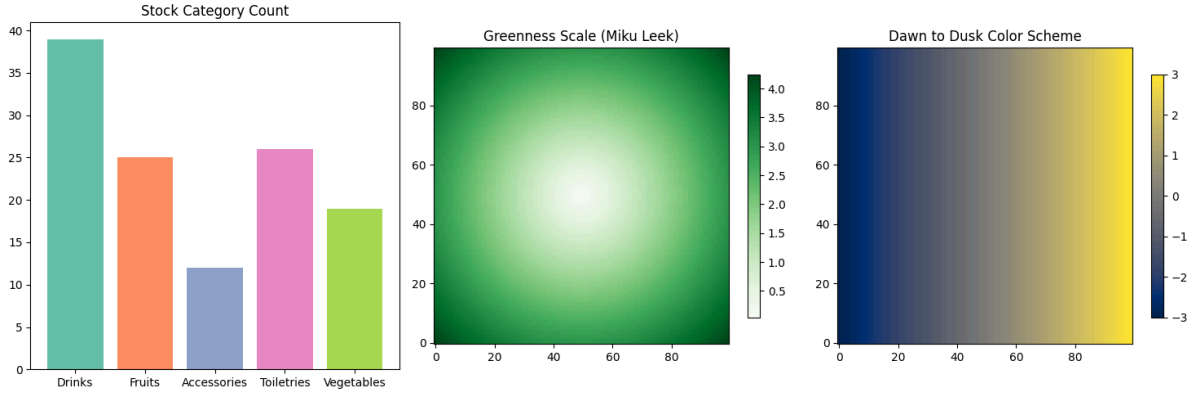
axes[0].bar(categories, values, color=plt.cm.Set2.colors[:len(categories)])
axes[0].set_title("Stock Category Count")

seq = axes[1].imshow(Z_seq, cmap="Greens", origin="lower")
axes[1].set_title("Greenness Scale (Miku Leek)")
plt.colorbar(seq, ax=axes[1], shrink=0.7)

div = axes[2].imshow(Z_div, cmap="cividis", origin="lower")
axes[2].set_title("Dawn to Dusk Color Scheme")
plt.colorbar(div, ax=axes[2], shrink=0.7)


plt.tight_layout()
plt.show()
```

Output Snippet



TASK 3 : CUSTOM COLOR USING THE HSL COLOR SYSTEM

Code Snippet

```python
import colorsys
import matplotlib.pyplot as plt

hues = [0, 65, 135, 200, 280]
s, l = 0.70, 0.50

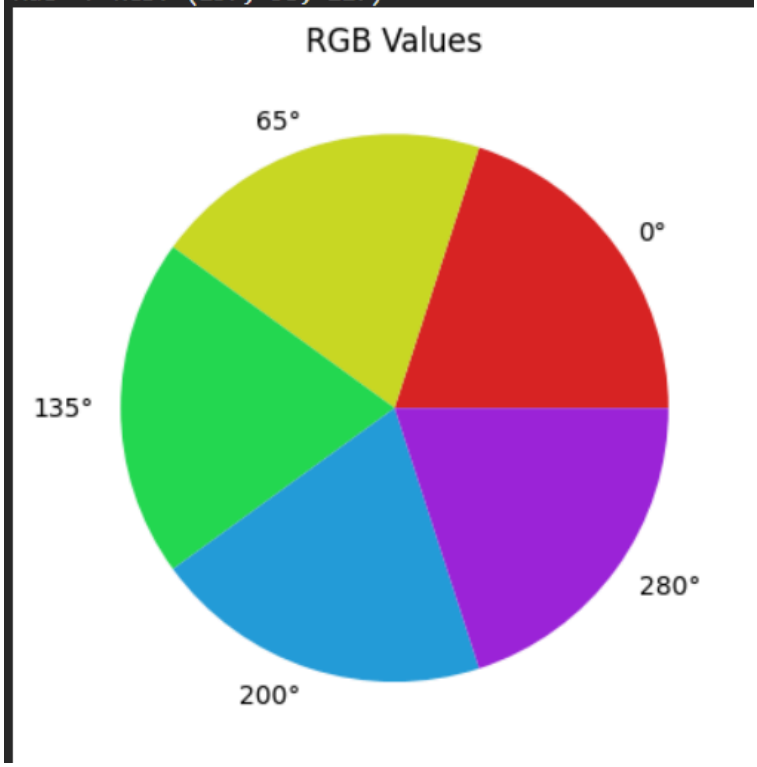
colors = [colorsys.hls_to_rgb(h/360, l, s) for h in hues]

for h, (r, g, b) in zip(hues, colors):
    print(f"Hue -> RGB: ({r*255:.0f}, {g*255:.0f}, {b*255:.0f})")

plt.title("RGB Values")
plt.pie([1]*5, colors=colors, labels=[f"{h}°" for h in hues])
plt.show()
```

Output Snippet

```
Hue -> RGB: (217, 38, 38)
Hue -> RGB: (202, 217, 38)
Hue -> RGB: (38, 217, 83)
Hue -> RGB: (38, 157, 217)
Hue -> RGB: (157, 38, 217)
```



TASK 4: COMMON MISTAKES IN THE USE OF COLOR IN DATA VISUALIZATION

Code Snippet

```python
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Sample data
categories = ["Passed", "Taking Summer Class", "Failed"]
values = [10, 25, 17]
categories2 = ["X", "Y", "Z"]
values2 = [23, 44, 37]

fig, axs = plt.subplots(3, 2, figsize=(12, 10))

axs[0,0].bar(categories, values, color=["pink", "orange", "purple"])
axs[0,0].set_title("Bad: Unnecessary Colors")

axs[0,1].bar(categories, values, color=["tab:green", "gold", "tab:red"])
axs[0,1].set_title("Good: Simple Colors")

colors = sns.color_palette("hsv", 12)
axs[1,0].bar(range(12), np.random.randint(5,20,12), color=colors)
axs[1,0].set_title("Bad: Excessive Colors")

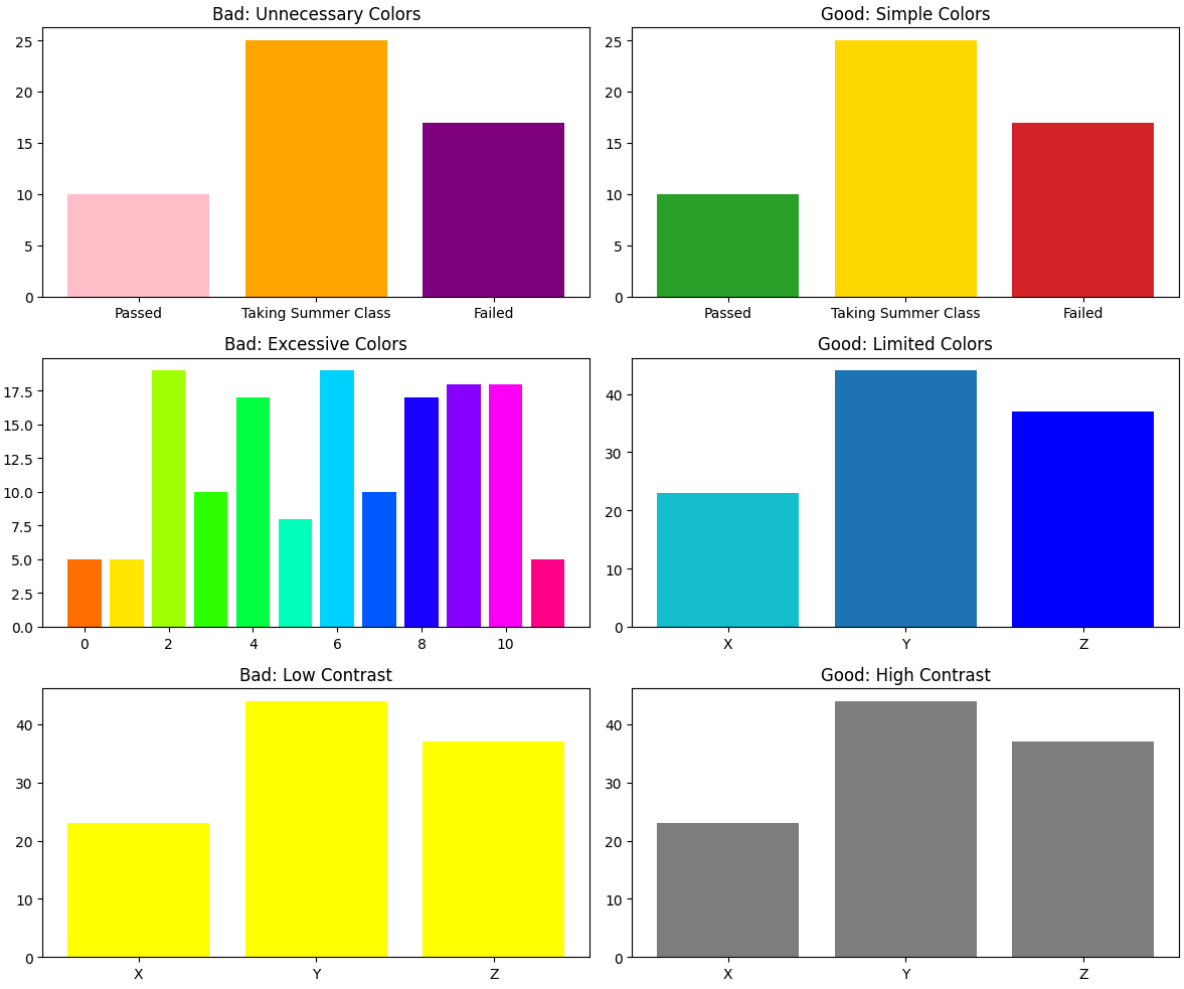
axs[1,1].bar(categories2, values2, color=["tab:cyan", "tab:blue", "blue"])
axs[1,1].set_title("Good: Limited Colors")

axs[2,0].bar(categories2, values2, color="yellow")
axs[2,0].set_facecolor("white")
axs[2,0].set_title("Bad: Low Contrast")

axs[2,1].bar(categories2, values2, color="gray")
axs[2,1].set_facecolor("white")
axs[2,1].set_title("Good: High Contrast")

plt.tight_layout()
plt.show()
```

Output Snippet



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Create a visualization project that demonstrates the purposeful use of color. Follow the steps below in one notebook/script.

1. Create a dataset

Example: 5 product categories with their sales values.

2. Color and Perception

Plot three bar charts: one showing hue differences, one showing saturation differences, and one showing luminance differences.

3. Color Schemes and Types of Data

Plot three more charts:

A categorical color scheme (each product = unique color).

A sequential color scheme (low → high values).

A diverging color scheme (below vs above average).

4. Custom Colors with HSL

Generate 5 custom HSL colors, convert them to RGB, and use them in a pie chart.

Print the RGB values of your chosen colors.

5. Common Mistakes and Corrections

Create one “bad” chart that uses too many random colors or poor contrast.

Create a corrected “good” version of the same chart with proper color use.

TASK 5 : SUPPLEMENTARY ACTIVITY

Code Snippet

Create a corrected "good" version of the same chart with proper color use.

```
import pandas as pd

url = "https://raw.githubusercontent.com/Pasioncpe/CPE-031-CPE2151---Visualizations-and-Data-Analysis---PASION/refs/heads/main/ev_market_2026.csv"

df = pd.read_csv(url)
```

```
df.head()
```

	brand	model	year	variant	price_usd	battery_capacity_kwh	range_miles	charging_speed_kw	acceleration_0_60_mph	top_speed_mph	...	body
0	Volkswagen	ID. Buzz	2023	Performance	104880.80	118.7	400.0	234.5	3.04	195.0	...	
1	Toyota	bZ Compact SUV	2022	Premium	48217.41	58.8	219.0	148.1	5.77	159.0	...	
2	GM/Chevrolet	Bolt EV	2024	Premium	49651.12	58.2	225.0	104.9	6.84	148.0	...	
3	Kia	Sportage EV	2024	Long Range	38131.56	102.5	349.0	66.5	4.66	176.0	...	
4	Tesla	Model 3	2022	Long Range	144079.87	93.9	314.0	298.5	5.64	165.0	...	

5 rows × 24 columns

```
df = df.drop(columns=['model', 'year', 'variant', 'price_usd', 'range_miles', 'charging_speed_kw', 'acceleration_0_60_mph', 'top_speed_mph', 'horsepow

df = df.head()
```

```
df_table = pd.DataFrame(df)
print(df_table)
x = df_table['brand']
y = df_table['battery_capacity_kwh']

fig, axs = plt.subplots(3, 3, figsize=(15,10))

# For number 2 (Hue, Saturation, Luminance demonstration from earlier task)
# Hue demonstration
axs[0,0].bar(x, y, color=['blue', 'tab:red', 'gold', 'gray', 'salmon'])
axs[0,0].set_title("TOP EV Brand Battery Life (kW/h) #1 (Hue)")

# Saturation demonstration
axs[0,1].bar(x, y, color=['#bfbfbf', '#baa4a2', '#b88b88', '#ba716c', '#b85d56
axs[0,1].set_title("TOP EV Brand Battery Life (kW/h) #1 (Saturation)")

# Luminance demonstration
axs[0,2].bar(x, y, color=['#000000', '#360300', '#730600', '#a80a02', '#db0b00
axs[0,2].set_title("TOP EV Brand Battery Life (kW/h) #1 (Luminance)")

# For number 3 (Color Schemes and Types of Data)
# Categorical Color Scheme
axs[1,0].bar(x, y, color=plt.cm.Set2.colors[:len(x)])
axs[1,0].set_title("Brand with Categorical Colors #2")

# Define variables for sequential and diverging plots
x_grid = np.linspace(-3, 3, 100)
y_grid = np.linspace(-3, 3, 100)
X_grid, Y_grid = np.meshgrid(x_grid, y_grid)
Z_seq = np.sqrt(X_grid**2 + Y_grid**2)
Z_div = X_grid

# Sequential Color Scheme
seq_plot = axs[1,1].imshow(Z_seq, cmap="Blues", origin="lower")
axs[1,1].set_title("Sequential Color Scheme #2")
fig.colorbar(seq_plot, ax=axs[1,1], shrink=0.7)

# Diverging Color Scheme
div_plot = axs[1,2].imshow(Z_div, cmap="coolwarm", origin="lower")
axs[1,2].set_title("Diverging Color Scheme #2")
fig.colorbar(div_plot, ax=axs[1,2], shrink=0.7)
```

```python
# PIE CHART
categories = ['Volkswagen', 'Toyota', 'GM/Chevrolet', 'Kia', 'Tesla']
values = [118.7, 58.8, 58.2, 102.5, 93.9]

hues = [int(i * 360 / len(values)) for i in range(len(values))]
s, l = 0.70, 0.50

colors = [colorsys.hls_to_rgb(h / 360, l, s) for h in hues]

for cat, h, (r, g, b) in zip(categories, hues, colors):
    print(f"{cat} (Hue {h}°) -> RGB: ({r*255:.0f}, {g*255:.0f}, {b*255:.0f})")

axs[2,0].set_title("RGB Values by EV Brand")
axs[2,0].pie(values,
    colors=colors,
    labels=[f"{cat} ({h}°)" for cat, h in zip(categories, hues)],
    autopct='%1.1f%%',
    startangle=90
)

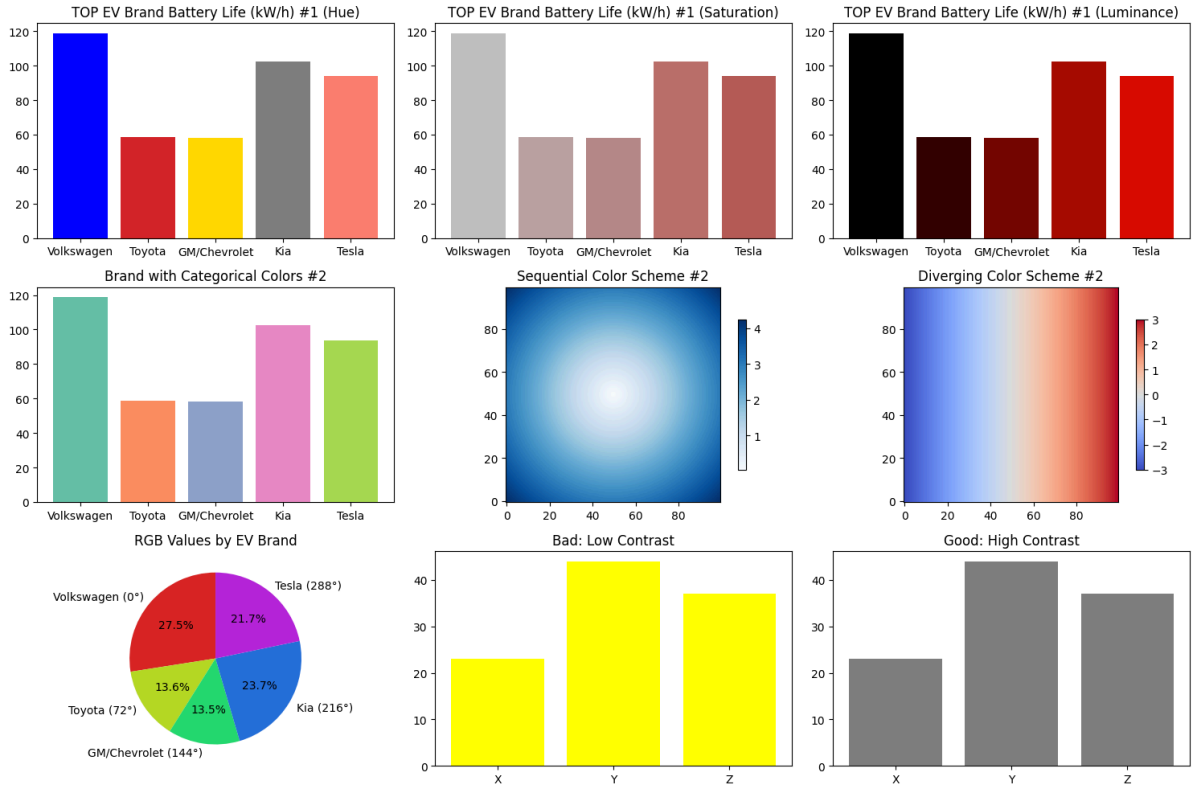
# Bad
axs[2,1].bar(categories2, values2, color="yellow")
axs[2,1].set_facecolor("white")
axs[2,1].set_title("Bad: Low Contrast")

# Good
axs[2,2].bar(categories2, values2, color="gray")
axs[2,2].set_facecolor("white")
axs[2,2].set_title("Good: High Contrast")


plt.tight_layout()
plt.show()
```

Output Snippet

```
          brand  battery_capacity_kwh
0    Volkswagen                 118.7
1        Toyota                  58.8
2  GM/Chevrolet                  58.2
3           Kia                 102.5
4         Tesla                  93.9
Volkswagen (Hue 0°) -> RGB: (217, 38, 38)
Toyota (Hue 72°) -> RGB: (181, 217, 38)
GM/Chevrolet (Hue 144°) -> RGB: (38, 217, 110)
Kia (Hue 216°) -> RGB: (38, 110, 217)
Tesla (Hue 288°) -> RGB: (181, 38, 217)
```



PANGILINAN CONCLUSION AND ANALYSIS

Throughout this entire activity, we learned how to use color purposefully in data visualization to communicate information more effectively rather than just decoratively. Working through the tasks showed how hue, saturation, and luminance, they serve each role in it. The saturation conveys the intensity of the color and luminance for the lightness and darkness that creates a contrast that guides our eyes toward the chart. As for the hue helps the categories apart. We also learned when to use different color schemes: categorical colors for separate groups, sequential colors for values that go from low to high, and diverging colors for values that go above or below an average. Using the HSL color system also helped us understand how colors can be created and converted into RGB values through code. Lastly, comparing the "bad" and "good" charts showed us that too many colors or low contrast can make a graph hard to read, while simple and high-contrast colors make the data much clearer.

As for the supplementary we use the datasets of electric vehicles, we made several charts showing the hue, saturation, and luminance. We also tried out different color schemes and 6 charts with one color per brand for categories, a color scheme going from light to dark for low to high values, and a scheme showing values above or below average. We picked our own colors using HSL and turned them into RGB to color a pie chart. Lastly, we compared a low-contrast yellow chart with a high-contrast gray chart, which showed that good contrast makes data much easier to read.

PASION CONCLUSION AND ANALYSIS

In conclusion, me and my groupmates learned basic color theory and how we can apply this into making effective charts/graphs for data visualization. We learned that Hue, Saturation, and Luminance are factors that can direct the eyes of the viewer. This can influence what a person may think or feel about the chart, so it is important to pick the right Hue, amount of saturation, and the luminance of the color/s. This concept also applies to the contrast, consistency, accessibility, and data-ink efficiency of the color. Details like “blending in the background”, “too many random colors”, “colors are not seen” and “colors don’t match categories” can affect the presentation of the information you intend to show to the viewers. Through this activity, we got to experiment and make different kinds of charts (bars/pies) with the goal of using color as an effective tool in presenting information more cleanly.

For the supplementary activity, we were challenged to get a data set, show the differences of hue, saturation, and luminance, color schemes (categorical/sequential/diverging), custom HSL colors in a pie chart, and identifying the common mistakes in using color for charts/graphs. We applied our previous lessons into getting the dataset, getting rid of the values not needed, isolating, and applying it into the graphs itself. We chose a data set that shows the characteristics of the top Electric Vehicles, specifically the battery life. On number 2, we applied different hues, different saturations (use of hex color codes), and the luminance (use of hex color codes as well). For number 3, we applied the color schemes with the presets that came with the library of MatPlotLib. In the pie chart, we chose different values for the HSL values from a 360 degree spectrum. And lastly, the bad and good, we simply thought of what might make a graph look bad or “hard to look at”.

TORRES CONCLUSION AND ANALYSIS

In Conclusion, I’ve learned how to use the colors to perceive the users on the right category of the dataset by its HUEs, SATURATIONs, and LUMINESCENCE which makes the graph more understandable and feasible to the naked eye. HUE explains how the colors perceive the users by the naked eye and tell what color is it like the RGB (Red, Green, and Blue). SATURATION is the intensity or purity of the color if it is (Vivid Red or Faded Red). LUMINESCENCE is the Lightness and Darkness of the color scheme (Light Green or Dark Green). In the TASKs, I’ve learned how to provide the bar graphs and pie chart that has HUE, SATURATION, and LUMINESCENCE color schemes to provide visual aids to the viewers and also determine the Good and Bad Effects of using Color in the user’s perception in the graph itself.

In the Supplementary Activity, I’ve learned on how to familiarized myself with the creation of a complete 3×3 subplot matrix in Matplotlib for analyzing how the choice of colors and their accessibility affects the appearance of data plots. Using the EV battery capacity data (battery_capacity_kwh), I illustrated how the values of Hue, Saturation, and Luminance influence the charts’ look and managed to assign Categorical, Sequential (Blues), and Diverging (coolwarm) palettes to their corresponding data types. In addition, I was able to demonstrate the EV market brand values assigned to the RGB colors’ mathematical representation using the colorsys.hls_to_rgb

library function inside the pie chart. Finally, I proved that using the low-contrast yellow bars instead of high-contrast gray bars significantly impacts the readability of the charts, including the relationship between foreground and background colors.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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